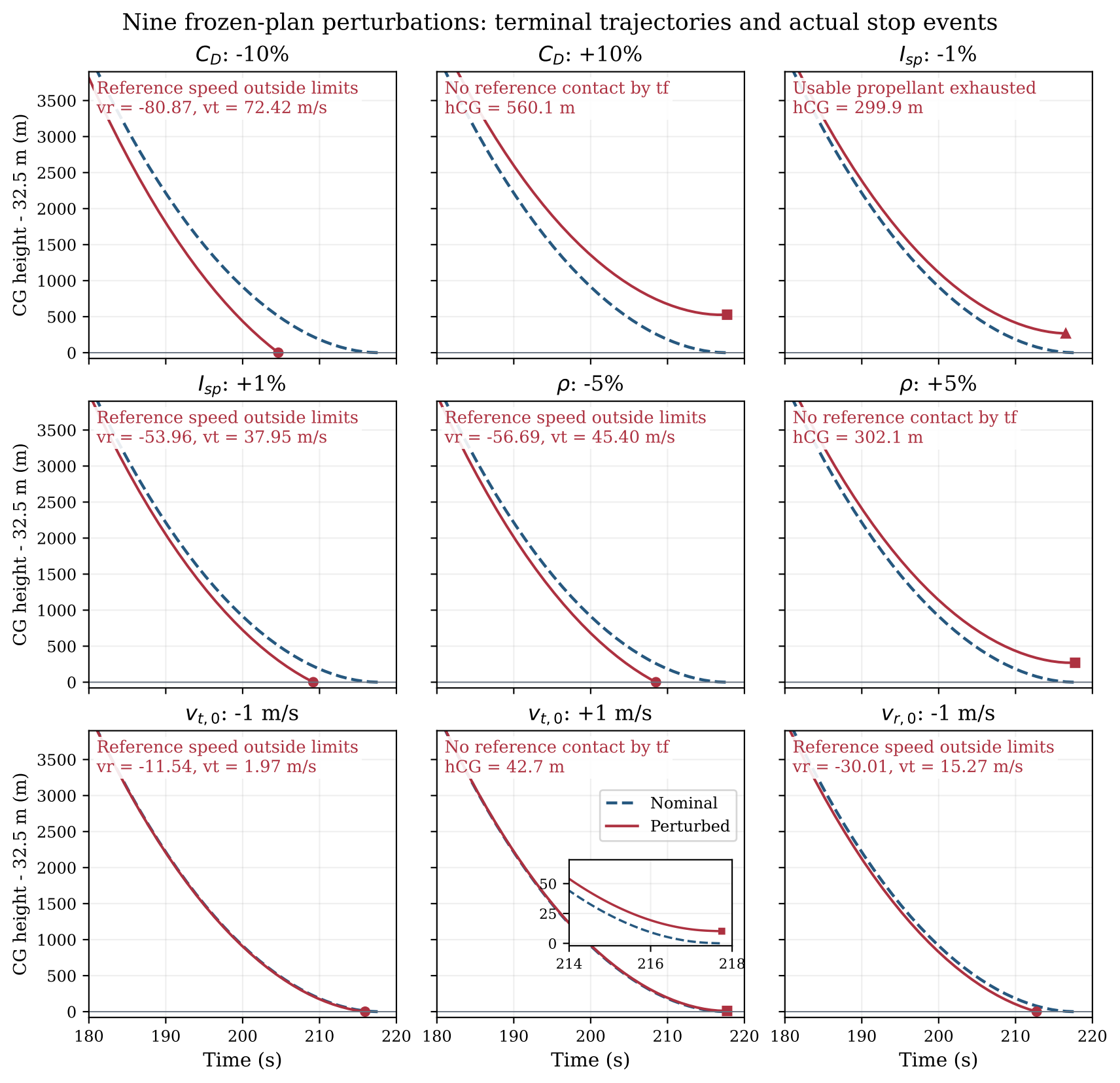
Nine frozen-plan perturbations: terminal trajectories and actual stop events
C_D: -10%
Reference speed outside limits
vr = -80.87, vt = 72.42 m/s
C_D: +10%
No reference contact by tf
hCG = 560.1 m
I_{sp}: -1%
Usable propellant exhausted
hCG = 299.9 m
I_{sp}: +1%
Reference speed outside limits
vr = -53.96, vt = 37.95 m/s
ρ: -5%
Reference speed outside limits
vr = -56.69, vt = 45.40 m/s
ρ: +5%
No reference contact by tf
hCG = 302.1 m
$v_{t,0}$: -1 m/s
Reference speed outside limits
vr = -11.54, vt = 1.97 m/s
$v_{t,0}$: +1 m/s
No reference contact by tf
hCG = 42.7 m
Nominal
Perturbed
$v_{r,0}$: -1 m/s
Reference speed outside limits
vr = -30.01, vt = 15.27 m/s
CG height - 32.5 m (m)
Time (s)